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Research Paper

Perceptions and recommendations of multiparas and health-related professionals on appropriate birth intervals: A descriptive qualitative study

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ABSTRACT

Objectives: This study aimed to explore the perceptions and recommendations of multiparas and health-related professionals regarding appropriate birth intervals (BIs) and key determinants.
Methods: In-depth semi-structured interviews were conducted between April 1 and June 30, 2022. Nine multiparas and thirteen health-related professionals were purposefully sampled until data saturation was reached. A thematic analysis approach was applied to the interview transcripts, utilizing dual independent coding and consensus validation in NVivo 12.0.
Results: The data generated two overarching categories: 1) balanced decision-making on the appropriate birth intervals and 2) internal and external determinants integrated with health and societal considerations. Four key themes emerged following the two categories: 1) consistency and discrepancy between the actual and recommended birth intervals of multiparas; 2) health- and development-oriented professional recommendations; 3) internal determinants related to individual-level factors; and 4) external determinants related to child-related factors, family support, and social security. Weighing women's reproductive health and career development, multiparas and health-related professionals perceived a length between 18 and 36 months as the appropriate BI.
Conclusion: Multiparas and health-related professionals shaped their balanced recommendations on a relatively appropriate birth interval ranging from 18 to 36 months, which was influenced by women's individual-level factors, child-related factors, family support, and social security. Targeted social and healthcare services should be offered to women and their families during the BIs.
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What is known?

- Existing research indicates that short and long birth intervals (BIs) are associated with increased risks of adverse maternal and child health outcomes.
- Despite WHO's evidence-based guidance advocating a minimum 24-month interval after a live birth, significant variability in observed BIs persists across populations.

What is new?

- This study identifies an 18–36 month BI consensus among multiparas and health-related professionals in China, balancing biomedical imperatives with sociocultural realities.
- It presents a novel framework illustrating the interplay of health and societal factors, including individual health and

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development, child-related considerations, family support, and social security, in shaping perceptions and recommendations regarding BIs within the Chinese context.
- The study highlights the alignment and discrepancies between actual and recommended BIs, revealing the complex negotiation between individual circumstances and sociocultural influences in reproductive planning.

1. Introduction

The birth interval (BI), the period between live birth and the conception date of the subsequent pregnancy, is critical for assessing adequate population birth density [1]. An appropriate BI significantly influences both maternal and child health [2]. Substantial evidence links both short and long BIs to increased risks of adverse health events. Shorter intervals are associated with elevated risks of preterm birth, premature membrane rupture, gestational diabetes, pregnancy-related hypertension, anemia, third-trimester bleeding, and maternal mortality [3–5], while longer intervals may contribute to a higher incidence of gestational diabetes, gestational hypertension, preeclampsia, eclampsia, premature rupture of membranes, uterine fibroids, hypothyroidism during pregnancy, dystocia, maternal blood transfusion, and obesity in the first and second trimesters [6–10]. In response to these risks, the WHO recommends a waiting period of 24 months after a live birth before attempting the next conception [1]. Despite this guidance, significant variability in observed BIs persists across different populations and individual circumstances [11].

Although evidence-based BI recommendations, such as those from the WHO, provide valuable benchmarks, their applicability within China's unique sociocultural and healthcare context warrants critical examination [12]. Chinese women navigate distinctive influences on postpartum recovery and subsequent pregnancy planning that international guidelines may not fully address. Culturally, strong familial expectations regarding childbirth timing and parity often exist, sometimes without commensurate emotional or practical support from partners or extended family networks [13]. Psychologically, women frequently experience considerable pressure in balancing maternal duties with occupational responsibilities and other caregiving roles, creating stressors distinct from those documented in Western settings [14]. Physiologically, maternal recovery trajectories can be significantly shaped by traditional postpartum practices (e.g., *Zuo Yue Zi* confinement) and variable access to continuous, integrated reproductive healthcare services [15]. These context-specific factors, including familial pressures, psychological stressors, culturally-mediated recovery norms, and healthcare accessibility, likely drive the patterns of BI in China diverging from international recommendations.

Critically, these intersecting influences manifest in distinctive decision-making dynamics around BI among Chinese multiparas. While variability in BI has been observed across many countries, many couples in China exhibit ambivalent attitudes toward pregnancy, characterized by both a desire for and hesitance about having additional children. Women frequently navigate competing familial pressures (e.g., spousal/parental expectations, first-child acceptance of siblings) alongside personal health considerations [16–18]. Traditional preferences, such as son preference, may further complicate reproductive timing decisions [19]. Simultaneously, the embodied experience of postpartum recovery mediated by cultural practices like *Zuo Yue Zi* creates physiological considerations distinct from Western contexts [15]. A literature review showed that educational attainment, employment status, and stable income influence reproductive decisions [20]. Additionally, age and the number of children born are crucial in explaining variations in individuals' willingness to reproduce [21]. Families may seek shorter birth intervals to have more children within a limited reproductive timeframe, especially if childbearing is delayed [22]. Conversely, career considerations and later marriage and conception may lengthen birth intervals [22,23]. These complex interplays between psychological burdens, kinship dynamics, and culturally framed bodily experiences generate BI determination processes not fully captured by existing international frameworks.

This context underscores the imperative to investigate BI perceptions grounded in lived realities. Understanding multiparas' experiential knowledge and health-related professionals' clinical insights is vital for developing culturally congruent and clinically feasible recommendations. Such insights are essential for optimizing family planning counseling and maternal healthcare services in China, aiming to improve women's reproductive health outcomes and well-being. Therefore, this study employed a descriptive qualitative design to explore multiparas' and health-related professionals' perceptions and recommendations regarding appropriate BIs and their determinants within the Chinese sociocultural landscape.

2. Methods

2.1. Study design

To gain a comprehensive understanding of the perceptions and recommendations of both multiparas and health-related professionals regarding appropriate BIs, a qualitative descriptive study incorporating semi-structured interviews and an interpretive dimension was used. Qualitative descriptive study typically is an eclectic but reasonable combination of sampling, data collection, analysis, and re-presentation techniques [24]. This approach enabled us to reveal the hidden meanings behind maternal and child health-related professionals and recipients of appropriate BIs.

2.2. Participants and recruitment

Purposive sampling was used to ensure the inclusion of diverse participants in the study environment [25]. The inclusion criteria for participating women were 1) aged between 18 and 49 years, 2) multiparas who had experienced giving birth to two children, and 3) multiparas who lived in Shanghai as permanent residents. Health-related professionals were eligible to participate if they met the following criteria: 1) maternal and child health professionals (obstetricians, pediatricians, midwives, obstetric nurses) and professionals in the field of health sociology, and 2) with more than one year of work experience. This purposive sampling method ensured that participants with relevant experiences and expertise were included, thus offering a detailed understanding of the factors influencing BIs. We targeted women who had given birth to at least two children to capture their first-hand perspectives on ideal BIs based on their experiences. Health-related professionals were chosen for their specialized knowledge of maternal and child health, providing a comprehensive view of BIs from clinical and sociological perspectives. This approach allowed an in-depth exploration of how these factors interact with and affect BIs recommendations. The sample size was determined by data saturation (that is when no new information was observed in the participants' experiences) [26], which was achieved after 22 participants (nine multiparas and 13 health-related professionals) had been interviewed. The participants belonged to various

occupational categories to achieve maximum diversity. All participants remained anonymous during the study period.

2.3. Data collection

To ensure a serene and comfortable environment, interviews were conducted in a small meeting room of the outpatient clinics at the Obstetrics and Gynecology Hospital in Shanghai, China. This hospital is a tertiary specialized institution affiliated with Fudan University, dedicated to women's health, and comprises two campuses (Huangpu campus and Yangpu campus). The study was conducted on the Huangpu campus between April 1, 2022, and June 30, 2022. Three research investigators (H. Liu, S. Liang, and X. Wang) conducted interviews with the 22 participants. Each interview was tape-recorded and commenced with an open-ended, thought-provoking question that encouraged participants to share their experiences and insights freely.

To ensure consistency across interviews, we strictly adhered to a semi-structured interview guide developed following established qualitative methodologies [27]. This guide comprised two distinct sets of questions tailored to the participant groups.

Set one, designed for multiparas, explored the following domains: factors influencing their personal BI decisions, considerations they recommended for reproductive-aged couples, perceived optimal BIs durations for women with varying health and developmental conditions (along with rationales), and the advantages and disadvantages of BIs lengths from the perspectives of career and family relationships. Specifically, the open-ended questions included: 1) What factors influenced your decision-making on your BIs? 2) What factors should be considered when planning the BIs for couples of reproductive age? 3) What is the most appropriate BIs for women with different health and developmental conditions, and why? 4) From the perspectives of women's individual career development and family relationships, what are the advantages or disadvantages of having either a too-long or short BIs period?

Set two, designed for health-related professionals, examined the following domains: clinical factors considered when determining appropriate BIs, recommended BI durations for diverse health and developmental conditions (with evidence-based justifications), and the implications of BI lengths on women's health, career trajectories, and social security systems. Specifically, the open-ended interview questions included: 1) What factors should be considered when determining an appropriate BI period? 2) What is the most appropriate BI for women with varying health and developmental conditions, and why? 3) From the perspectives of women's health, career development, and social security, what are the advantages or disadvantages of having either a too-long or short BIs period?

This dual-perspective approach facilitated the triangulation of lived experiences with clinical expertise, aligning with best practices in qualitative research. The interviewers diligently noted and documented the participants' expressions, emotions, and behaviors, capturing valuable qualitative data. Each interview typically lasted 40–60 min and was conducted in Mandarin, with on-the-spot recordings to facilitate data analysis.

2.4. Ethics approval

Ethical approval was obtained from the School of Public Health Ethics Committee at Fudan University (Approval ID: IRB#2021-03-0880). Before each interview, all potential participants received an informative document outlining the purpose of the study. Participants were explicitly informed of their right to decline participation and the option to withdraw from the study at any point after agreeing to participate. Those willing to participate were required to provide verbal agreement and sign an informed consent form.

2.5. Data analysis

We adopted thematic analysis based on the approach of Braun et al. [28] to identify key themes from the interview data. All audio-recorded interviews were transcribed verbatim in Chinese and repeatedly read to gain a comprehensive understanding of participants' perceptions and experiences. The data were first segmented into smaller meaningful units and labeled with codes that reflected factors influencing appropriate birth intervals. The research team compared and contrasted the codes, grouped them into categories and iteratively refined them through the ongoing analysis. This inductive process continued until data saturation was reached, meaning no new themes emerged. To manage and facilitate the analysis process, NVivo 12.0 software was used. This helped systematically store, retrieve, and visualize the coded data and themes throughout the research process.

2.6. Trustworthiness

To ensure trustworthiness in this qualitative study, we adhered to the criteria of credibility, transferability, dependability, and confirmability [29]. Interviews were conducted empathetically and without coercion to encourage open responses from participants. Credibility was further enhanced through member checking, where interview transcripts were returned to participants to validate accuracy and interpretive authenticity. Transferability was supported by purposive sampling of participants from diverse backgrounds, including multiparas and various health-related professionals, to strengthen the applicability of findings to similar contexts. For dependability and confirmability, two researchers (H. Liu and S. Liang) independently performed initial coding and thematic development, with iterative revisions emerging from consensus discussions. The analytical process was reviewed by a third researcher (X. Wang) to reduce subjective interpretation and enhance neutrality. Researchers bracketed their preconceptions and maintained detailed audit trails of decision-making throughout the study. The cross-linguistic validation of Chinese quotations was rigorously executed: findings were initially translated into English, then back-translated into Chinese, with the research team's final verification of semantic equivalence. Data-theme coherence was continuously ensured through repeated comparative analysis.

3. Results

The average age of multiparas was 33.22 $\pm$ 4.92 years. The interviewed health-related professionals were aged 26–59 years. Appendixes A and B present the sample characteristics of the interviewed multiparas and health-related professionals. The data generated two overarching categories: 1) balanced decision-making on the appropriate BIs and 2) internal and external factors integrated with health and societal considerations. Four key themes emerged following the two categories: 1) consistency and discrepancy between the actual and recommended BIs of multiparas; 2) health- and development-oriented professional recommendations; 3) internal determinants related to individual-level factors; and 4) external determinants of child-related factors, family support, and social security. Weighing women's reproductive health and career development, multiparas and health-related professionals perceived a length of 18–36 months as the appropriate BI(Appendix C).

3.1. Category 1: Balanced decision-making on the appropriate BI

In this section, we describe the decision-making processes of multiparas and health-related professionals regarding their recommendations on appropriate BIs, which is a balanced decision weighing the health and career development of women. Of the nine participating women, five recommended an appropriate BI of 18–36 months; two recommended a length of >3 years, and the other two recommended less than 18 months. Of the 13 health-related professionals, eight recommended an appropriate BI of 18–36 months, four recommended a length of >36 months, and one recommended <18 months.

3.1.1. Consistency and discrepancy between the actual and recommended BIs of multiparas

The nine interviewed multiparas reported their actual length and recommended appropriate length of BIs (Appendix A). Most women expressed a consistent attitude toward the actual and recommended BIs. In contrast, the other four women made a discrepant decision on the recommended appropriate BIs, either longer or shorter than their actual BIs. Women's individual preferences, health status, support from partners and family caregivers, and consideration of their job issues may constitute the primary functional elements explaining the consistency and discrepancy effects of the actual and recommended BIs of women.

> *"The interval between my first and second baby was 21 months. This interval is OK for me, and I would also recommend it to other women like me…it is not so long, and my parents would like to help me take care of my kids…they are still energetic enough. I think the interval between 18 and 36 months is good and acceptable."* (W05)

> *"My birth interval is 128 months, which is too long. I had planned to have a second baby within less than this interval…but at that time, my mother and my mother-in-law were not in good health, and also I had just changed my jobs. So we had to wait and plan for another baby at the proper time. For other women, I would say that the interval between 18 and 24 months is appropriate."* (W08)

3.1.2. Health- and development-oriented professional recommendations

The interviewed health-related professionals also made recommendations on the appropriate BIs in consideration of both the health and well-being of the women themselves and their present and future career development. Most participating professionals consistently recommended that a BI ranging from 18 to 36 months should be an appropriate interval for women of reproductive age (Appendix B). Maternal health- and development-oriented decision-making on recommended BI became a fundamental trade-off, which allowed the woman sufficient time to recover and maintain good health and thoroughly plan her career development, education, and promotion opportunities.

> *"An appropriate BI is important in promoting the woman and her child's health. From this perspective, I would recommend an appropriate interval of 18 and 36 months. On the other hand, focusing solely on the woman's career path, or if she plans to continue further education, I would suggest extending the interval to some degree, like between 24 and 60 months…Of course, in reality, we need to balance the two aspects, and I would say that the interval ranging from 18 to 36 months will be reasonable."* (HP07)

3.2. Caregory 2: Internal and external determinants integrated with health and societal considerations

In this section, we describe the key determinants affecting the decisions of multiparas and health-related professionals regarding appropriate BIs. They shaped their decision-making based on health and societal considerations. Specific internal and external factors play interactive roles in determining or recommending appropriate BIs for women of reproductive age. These factors primarily consist of individual-level factors, child-related factors, family support, and social security.

3.2.1. Individual-level factors

All nine interviewed multiparas who had given birth to two children expressed that the BIs were primarily influenced by several internal determinants related to individual-level factors, including postpartum recovery status, previous childbirth experiences, reproductive age, and career development plans. These determinants are significant personal factors affecting women's decision-making regarding BIs.

> *"I thought that I had almost recovered. I was already 31 when my husband and I planned our second child…So we had to 'hurry up' to have the second baby"* (W02)

> *"The delivery of my first child went very smoothly. Usually, I am not too busy with my job, so during my pregnancies, I continued working until the onset of labor…I always thought about having two children."* (W03)

> *"I had my first child at thirty…that was not young. So, I planned to have a second child before forty…A woman who gets pregnant after 40 years old will have more health risks due to advanced age. A BI between 3 and 5 years would be appropriate for women like me."* (W07)

The interviewed health-related professionals pointed out that various factors, including maternal age, the health status of women, future career development, previous pregnancy, and childbirth experiences, mode of delivery, and social network relationships, can determine an appropriate BI. These factors constitute the fundamental elements that health-related professionals consider when recommending an appropriate BI for women of reproductive age.

> *"For a woman at a younger age, the BI length is not so crucial. If the woman's age at the previous pregnancy was close to advanced maternal age, a shorter interval would be better when she wants a second child."* (HP01)

> *"If the woman had a positive experience during her previous pregnancy and childbirth, and her last delivery went well and was natural, she would often recover faster. This might encourage her to plan for a second baby sooner."* (HP04)

> *"From the perspective of career development, the longer the BI, the better. However, we do not recommend the interval exceeding 10 years or above the criteria for advanced maternal age."* (HP12)

3.2.2. Child-related factors

The women interviewed in this study mentioned that their

family planning decisions were influenced by several factors, including the age of their first child, childcare costs, and caregiving capabilities. For instance, given that their first child was still young, parents felt compelled to dedicate more time and effort to their care. They also expressed concerns that the arrival of a second child might affect the existing parent-child relationship. Some participants reported that they did not intend to conceive a second child until their first child had matured and developed a certain degree of self-sufficiency. In addition, some women were influenced by traditional and regional cultural norms, which, alongside the sex of their children, affected the timing of subsequent births.

"At that time, the main concern was that my first child was too young. If a second child was to be born soon, taking good care of both kids might be challenging. When the first child was around three years old, we started to plan for the second child." (W07)

"I'm a Shanghainese, and my husband is from Shandong, where people often want boys…Our first kid is a girl…So when the first child grew older, and we didn't have to spend all day with her, my husband and I began considering having a second child. He would prefer a boy." (W05)

The health-related professionals indicated that parents would gradually pay less attention to their children as their independence increased at three. Most participating professionals recommended an appropriate BI of 1.5–3 years due to breastfeeding cessation and the change in the feeding modes of children, the growth and development of the body movements of children, and their decreased care needs.

"Infants in their first year are vulnerable and susceptible to danger and risk, so parents have to spend a lot of time and energy in taking care of them. But it will be much easier for parents to watch over them when they get older. This will allow the mother to plan for the subsequent pregnancies." (HP10)

"When thinking about having a second baby, many women wait until they stop breastfeeding their first child and switch to other feeding methods. Usually, their first kid starts kindergarten around the age of three. That's a good time for the family to plan for a second child. So, a three-year gap seems pretty reasonable." (HP06)

"Two kids in the family can be playmates with each other. Besides, having two children is beneficial for learning and education for the kids. Such a family environment will help promote the overall growth and development of the children." (HP13)

3.2.3. Family support

A supportive family environment, economic conditions, the reproductive experiences of relatives, and the age of their spouse may influence the choices and recommendations of women for appropriate BIs. In addition, most participating women believed that caregiving and emotional support from family members and their assistance in childcare were influential external factors in their decisions to have a second child.

"We live together with my parents, and they also provide us with financial assistance. Thus, we don't have much pressure to have a second child." (W01)

"Being a dual-earner family, I think it would be challenging to take care of two kids without the help of the grandparents. Fortunately, my parents are very supportive, and they've been helping us look after the two children." (W09)

The interviewed health-related professionals also stated that family support was an essential factor to consider when providing advice on fertility plans for women of childbearing age. As the cost of raising two children gradually increased, families attempted to build a stable economic foundation to support their children's lives and educational expenses.

"The cost of child care should be considered for many families. Most families plan for the next child under stable and mature economic conditions. Thus, financial support is critical." (HP7)

"The young couples might consider postponing their plan for next pregnancy if the grandparents can't help them take care of the kids…Or when their economic foundation is firm enough to hire a babysitter, they would then make the plan for a second or a third child." (S11)

3.2.4. Social security

The interviewed women described an external supportive social environment as another consideration for their decision-making regarding appropriate BIs. This environment includes good social and medical systems, maternity security policies, and childcare services, which can enhance the willingness of women to make family plans and select proper birth intervals.

"Early childcare settings in our country are quite underdeveloped. Instead, kids at tiny months of age can be arranged in full-time care in the childcare settings of developed countries…So it seems to us that considering another child in the coming one to two years is challenging and impractical." (W08)

"For elderly grandparents, taking care of two grandchildren becomes burdensome. After all, it takes a lot of time and energy for them to look after a toddler aged one or two. The social support for childcare services is not well established…So we can't afford to have a second child too soon." (W04)

"At first, according to China's one-child policy, I did not meet the criteria for having a second child. Then, when the new birth policy was implemented, we did want more kids in our family, and then we just made the plan of having a second child." (W09)

Health-related professionals mentioned that the support of the expansion of childcare institutions could improve the willingness of women to become fertile and shorten their BIs. The current policy inclines toward women of reproductive age by providing prolonged and paid maternity leave and encourages organizations to create a harmonious reproductive environment and career opportunities for women. This contributes to the occupational stability of women and stimulates their fertility intentions.

"According to the relevant national policies, the duration of maternity leave has been extended to about 6 months postpartum, allowing women to invest more time in children's caregiving. This policy aims to encourage couples to have more children. So, an interval of about 2-3 years would be reasonable for women with the fertility intention of two or more children." (HP02)

"The government and society should create a more flexible and favorable environment for appropriate BIs. For example, to ensure the stability of women's occupation and to develop more suitable job types for caring for two or more children…I think this would increase women's fertility intention of having more kids. Besides, creating a better social space, providing socialized parenting support, and utilizing the community efforts could be effective

ways to provide social support and guarantee for mothers and their family members." (HP13)

4. Discussion

Through a cultural lens, this qualitative study explored perceptions and recommendations regarding appropriate BIs and key determinants among multiparas and health-related professionals in China. Our findings reveal that BI recommendations of 18–36 months reflect a characteristic negotiation between reproductive health requirements and sociocultural imperatives unique to China.

In the present study, we found that the median length of the actual BI for multiparas was 21 months, with the majority of interviewed multiparas and health-related professionals agreeing on the appropriateness of BIs within the 18–36 months range. The actual BIs of women fell within the recommended range, aligning with the WHO's guideline for spacing after a live birth [1]. To reduce the risk of adverse maternal, perinatal, and infant health outcomes, the WHO recommends at least 24 months before attempting the subsequent pregnancy. This recommendation represents a compromise between two expert groups: one advocating for an optimal interval of 18 months and the other suggesting that evidence supports a recommended interval of 27 months [1]. In contrast, the recommended BIs in our study are balanced and justified by considerations of both health and development for women. When choosing the timing of the subsequent pregnancy, individuals and couples should weigh health risks and benefits alongside their age, fertility aspirations, access to health services, child-rearing support, and personal preferences [30]. Thus, the recommended appropriate BIs range in our study addresses women's reproductive health needs while facilitating their return to career development.

This study reveals that perceptions of appropriate BIs in China are fundamentally shaped by unique cultural norms that interact with contemporary socioeconomic transitions. These perceptions are influenced by multiple determinants, including individual-level factors, child-related considerations, family support systems, and social security provisions, as evidenced by multiparas' and health-related professionals' recommendations. Individual-level factors primarily comprised reproductive age, personal health status, previous childbirth experiences, and the future developmental needs of women. Women aged 35 years and older are likely to desire a second child promptly, possibly due to the awareness that advancing age can lead to infertility and an increased risk of pregnancy-related complications [31]. Conversely, other studies have shown that compared to younger women, older women may experience a decline in fertility and a reduced likelihood of pregnancy, thus leading to longer BIs [32,33]. Notably, participants in our study reported that women's health status is a significant personal factor affecting their recommendations for appropriate BIs, aligning with the viewpoints of Mishra et al. [34]. Beyond that, the developmental needs of women, including career development, educational opportunities, and promotion prospects, can also influence their decision-making regarding BIs. Due to the inevitable time and energy demands of pregnancy, childbirth, and child-rearing, the opportunity cost of having more children is higher for professional women than those engaged in household work [35]. Zhang et al. [4] also report that longer BIs are associated with a reduced likelihood of occupational interruptions in women, which supports female labor force participation.

In our study, child-related factors, such as the age and sex of the first child, changes in children's feeding modes, and childcare costs were perceived by multiparas and health-related professionals as essential determinants of appropriate recommended BIs. The majority of participants recommended a suitable BI of 1.5–3 years due to breastfeeding cessation, the physical and mental growth and development of children, and the decreased caregiving needs as children age. When the first child is between 3 and 6 years old, couples tend to have a higher intention to plan for a second child [36]. This trend may be attributed to the fact that children can attend kindergarten at three years old and begin formal schooling after six years old, thus allowing couples more time and energy to consider nurturing another child. In contrast, persistent patrilineal values continue to shorten BIs, particularly when firstborns are female. Despite urbanization, son preference remains a potent cultural driver. The female sex of the previous child and shorter breastfeeding duration were consistently associated with shorter BIs [37]. This highlights the importance and necessity of promoting breastfeeding and addressing preferences for male children, which presents a complex and long-term challenge.

Concurrently, we determined that a robust family support system, including supportive family members, stable economic conditions, and spouses' ages, can influence decision-making and recommendations regarding appropriate BIs, consistent with findings from other studies [3,30,38]. Chinese decisions regarding BIs operate within a familial collectivism paradigm distinct from individualistic models. Additionally, inadequate childcare infrastructure amplifies reliance on multigenerational support systems [39,40]. When grandparents are unavailable or unwilling to provide full-time care, BIs are extended regardless of maternal preferences, transforming cultural expectations into actual constraints. This dependency relationship reflects a uniquely Chinese adaptation to limited public services. Conversely, a supportive family environment can mitigate work-family conflicts, resulting in a significant reduction in the BIs for women having their second child. Moreover, economic and child-rearing barriers impede having one more child [41,42]. Over the past few decades, childbearing at later ages has become increasingly prevalent. Similar to women of advanced maternal age, those with advanced paternal age are associated with an increased risk of adverse reproductive outcomes, including spontaneous miscarriage [43]. Therefore, family factors related to appropriate BIs may also have implications for interpregnancy care or preconception counseling for couples planning their next child.

Embedded within China's unique sociocultural fabric, our findings reveal how social security systems reconfigure BI decisions. A well-structured and supportive system, which includes extended and paid maternal/parental leave, as well as accessible childcare services, can significantly enhance women's willingness to have children and to plan appropriate BIs. Although the traditional emphasis on multigenerational continuity creates inherent pressure for shorter BIs, contemporary realities—particularly women's dual roles as caregivers and economic providers—necessitate institutional support. Paid paternal leave is a key enabler of employment stability, allowing both parents to participate in early childcare actively. This measure has also been linked to a meaningful rise in second and third-birth rates [8]. Additionally, the women interviewed in our study expressed a strong need for expanded childcare services when planning for an appropriate BI. Paid maternal or parental leave is an essential bridge between childbirth and the start of available childcare services while promoting a more equitable division of caregiving responsibilities between genders [44]. Overall, a continuous and well-integrated social security framework, achieved through culturally attuned integration of state policy and indigenous traditions, can help mitigate financial and temporal constraints associated with raising multiple children. To support couples in

making informed decisions about appropriate BIs, governments should strengthen sustainable, family-friendly policies that alleviate child-rearing pressures, particularly by enhancing paid leave provisions and culturally competent childcare support.

5. Limitations

This study has several limitations that should be considered when interpreting the findings. First, the research was conducted exclusively at a tertiary women's hospital in urban China, with participation limited to multiparas who had delivered two singleton children. Notably excluded were primiparas, multiparous women whose second delivery involved a non-singleton birth, mothers undergoing a third delivery, and those planning a third pregnancy. Second, as most participants had recently given birth at the study site, the findings may have limited generalizability to other healthcare settings or mothers with older children. Additionally, the study population consisted primarily of long-term Shanghai residents with relatively high social support. The interviewed health professionals did not include reproductive medicine specialists or health economists. Accordingly, these findings, derived from a relatively limited sample, may not fully represent or apply to broader populations in other regions. We therefore recommend cautious interpretation and application of these results to different contexts.

6. Conclusions

This qualitative study illuminates culturally embedded perceptions of appropriate BIs among multiparas and health-related professionals in China. The consensus recommendation of 18–36 months reflects a distinctly Chinese negotiation between reproductive health imperatives and sociocultural realities. Variations in BIs illustrate diverse reproductive patterns and evolving societal norms. Key findings indicate that the determination of BIs operates within a family-centered decision-making framework, where multigenerational expectations, child-related factors, individual reproductive autonomy, and institutional support systems dynamically impact appropriate intervals. Therefore, clinical practice should integrate medical, psychological, and sociocultural competencies into BI counseling protocols. Policy innovations should develop culturally attuned support systems, including community childcare cooperatives and mechanisms that protect women's career continuity. Enhancing postpartum care and family planning services is essential for supporting women during the BI period. Collaborative efforts are necessary to foster an enabling environment for reproductive-aged couples. Further research is recommended to investigate the longitudinal effects of BI variations within China's unique cultural context. Fostering appropriate BIs in China requires transcending biomedical paradigms to embrace culturally grounded reproductive support models that harmonize tradition with modernity.

Data availability statement

The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Hongyan Liu: Conceptualization, Investigation, Writing – original draft, Data curation. **Xinli Zhu:** Data curation, Investigation, Validation. **Zhijing Xu:** Supervision. **Shuang Liang:** Data curation, Software, Visualization. **Ya Liu:** Data curation, Investigation. **Xiaojiao Wang:** Investigation, Software, Data curation, Visualization. **Xu Qian:** Conceptualization, Supervision, Formal analysis. **Chunyi Gu:** Conceptualization, Methodology, Writing – review & editing, Formal analysis, Project administration.

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Declaration of competing interest

The authors declare there is no conflict of interest.

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Appendices. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijnss.2025.07.001.

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